

BALASHOV, Russian FederationWMO: **341520**Lat: **51.550N**Lon: **43.150E**Elev: **159**StdP: **99.43**Time Zone: **4.00 (E04)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.4	-21.5	-27.3	0.3	-23.9	-24.6	0.4	-21.1	8.8	-5.5	8.1	-5.6	1.9	30	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	33.1	19.9	30.9	19.6	29.0	19.0	21.6	29.1	20.7	28.3	19.9	27.0	2.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	24.6	18.3	13.4	23.7	17.4	12.7	23.1	63.6	29.1	60.5	28.5	57.7	27.0	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.9	6.1	-28.6	34.8	4.5	2.4	-31.8	36.6	-34.5	38.0	-37.0	39.3	-40.3	41.1		
(5)			-28.8	24.2	4.4	1.8	-31.9	25.6	-34.5	26.6	-37.0	27.7	-40.2	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-8.0	-7.8	-2.2	8.1	15.1	19.2	21.8	20.1	14.2	7.1	-0.4	-5.9		
	DBStd	11.99	6.52	6.71	5.07	5.07	4.88	4.24	3.81	4.34	4.45	4.63	5.14	6.46		
	HDD10.0	2472	558	499	379	93	12	1	0	0	12	115	311	492		
	HDD18.3	4498	817	733	637	307	126	41	11	30	138	349	561	750		
	CDD10.0	1320	0	0	0	37	168	276	364	313	137	24	0	0		
	CDD18.3	304	0	0	0	1	24	66	117	85	13	0	0	0		
	CDH23.3	2931	0	0	0	11	234	598	1126	847	114	1	0	0		
	CDH26.7	1058	0	0	0	0	58	199	447	334	21	0	0	0		
(14) Wind	WSAvg	2.9	3.1	3.4	3.4	3.2	2.9	2.6	2.4	2.5	2.5	2.8	3.2	3.3		
(15) Precipitation	PrecAvg	500	39	29	28	32	38	55	57	46	42	41	52	46		
	PrecMax	746	84	68	67	71	111	153	188	130	116	120	158	83		
	PrecMin	333	3	0	0	7	2	8	17	0	0	1	4	17		
	PrecStd	97	19	17	16	18	26	35	35	31	32	26	31	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	2.9	12.5	24.8	30.6	34.3	36.0	36.3	29.0	21.2	12.3	7.1		
		MCWB	4.0	1.8	7.7	13.6	17.6	19.1	19.8	20.9	16.5	13.7	10.7	6.0		
	2%	DB	2.0	2.1	8.5	22.0	28.2	31.1	33.2	32.8	26.4	18.3	9.9	4.2		
		MCWB	1.3	1.3	5.8	12.6	16.9	19.3	20.3	19.8	16.2	12.5	8.4	3.6		
	5%	DB	1.0	1.3	6.1	19.0	26.1	28.8	31.2	30.3	24.2	15.9	8.1	2.5		
		MCWB	0.5	0.7	3.9	11.3	16.3	18.9	20.1	19.4	15.6	11.9	7.0	2.0		
	10%	DB	0.4	0.7	3.9	16.7	23.6	26.8	29.0	28.0	21.9	13.9	6.2	1.4		
		MCWB	-0.1	0.1	2.3	10.4	15.1	18.4	19.7	18.4	14.7	10.8	5.3	1.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	2.2	8.5	15.3	19.6	21.9	23.8	23.0	18.6	15.6	11.1	6.3		
		MCDB	4.6	2.5	10.8	22.3	27.3	28.8	29.2	29.7	24.2	18.5	12.0	6.8		
	2%	WB	1.6	1.5	6.3	13.6	18.1	20.7	21.9	21.2	17.4	13.9	8.9	3.6		
		MCDB	1.8	1.8	8.4	19.5	25.6	27.7	29.5	29.0	23.4	16.4	9.8	4.1		
	5%	WB	0.6	0.8	4.0	12.1	17.0	19.9	21.0	20.2	16.5	12.3	7.1	2.1		
		MCDB	0.9	1.3	5.8	17.7	24.4	26.7	28.9	28.0	22.3	15.0	7.9	2.4		
	10%	WB	0.0	0.2	2.4	10.8	15.9	19.0	20.2	19.2	15.3	11.0	5.4	1.1		
		MCDB	0.3	0.7	3.6	16.0	22.4	25.6	27.4	26.3	20.9	13.7	6.2	1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.2	6.7	9.6	10.4	9.9	10.3	10.9	10.0	7.5	5.1	5.1		
		MCDBR	3.7	4.3	8.6	13.4	12.8	12.3	12.7	13.4	13.1	10.8	7.0	3.8		
	5% WB	MCWBR	3.8	4.1	6.0	6.5	5.6	5.1	5.0	5.4	6.4	6.5	5.8	3.7		
		MCWBR	3.8	4.0	8.0	12.1	11.7	11.0	11.0	11.7	11.5	9.2	6.3	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.339	0.366	0.433	0.399	0.390	0.412	0.432	0.383	0.359	0.335	0.307		
	taud		2.107	2.013	2.064	2.102	2.307	2.360	2.295	2.224	2.343	2.351	2.263	2.138		
	Ebn at Noon		670	745	811	800	855	865	837	795	797	739	641	597		
	Edh at Noon		77	113	131	143	122	117	123	125	100	82	68	64		
(44) All-Sky Solar Radiation	RadAvg		0.92	1.97	3.41	4.32	5.73	6.08	6.06	5.16	3.48	1.81	0.77	0.51		
	RadStd		0.23	0.32	0.49	0.37	0.53	0.52	0.51	0.41	0.41	0.29	0.16	0.14		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (6 neighbors)	DBAvg	+0.97	N/A	N/A	+1.80	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page